Experiment 4: Dense Map Reconstruction and Map-Quality Evaluation

Raw write-up for Chapter 4, Section 4.4

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Reporting cutoff: 26 October 2026

Purpose of this document. This is the raw, complete write-up of Experiment 4, written before the thesis chapter; Section 4.4 of the final report is extracted from it. Every number traces to a file under `output/`, `map/` or `dataset/` named in the "Reporting rules" paragraph of its slot. Open concerns are set in red where they arise.

4.4.1 Simulation

Status: partial. Three databases are scored for geometry and drivability: the tiny warehouse, which passes, and two realistic-warehouse maps, one per layout, built on 26 October 2026 with loop closure deliberately disabled, which fail the quality criterion fixed before scoring. The tiny map's independent drivability reference remains lost.

Objective. Verify that the reconstructed map reflects the physical environment and is usable by a route planner.

Process. RTAB-Map 0.23.7 built a database from a mapping run. The occupancy grid was seated in world coordinates by one rigid transform measured from the map's own occupied cells against the world file, without odometry or ground truth. Geometry was then scored against the world's collision meshes, and drivability by stamping the true nav2 footprint along a ground-truth route.

Datasets. Table 4.1 lists the tiny-warehouse recordings and table 4.2 the realistic-warehouse ones. The tiny map's drivability reference is a separate run that no longer exists in either storage location; the only surviving recording of the tiny warehouse is the one that built the map. Each realistic map is probed with a recording that did not build it.

Table 4.1: Recordings of Experiment 4. Durations are rosbag wall-clock recording durations from the dataset registry; the simulator ran below real time.

Recording	World	Dur. (s)	Messages	Role
`dataset_map_tiny`	tiny, static	63.3	82 555	built `map/map_tiny.db`; substitute drivability route
`dataset_static_tiny_repeat`	tiny, static	–	2973 poses	independent drivability route; **lost**
`dataset_map_01`	large, static	193.5	152 291	large-warehouse mapping run; unscored

Table 4.2: Realistic-warehouse recordings of Experiment 4, all prefixed `dataset_allsensor_`. Build recordings are on the removable drive; their span is the stamp range of the stored nodes. Probe recordings are the static twins listed in the recordings table of the realistic-world note, read from the bags. U and I are the two layouts described there and in Chapter 3.

Recording	World	Span (s)	Truth	Role
`realistic_mapv2`	U, static	285.7	14 411	built `map/map_realistic`
`realistic_01_map`	I, static	314.9	15 760	built `map/map_realistic_01`
`realistic_000`	U, static	125.5	6 278	independent probe, 171.3 m
`realistic_001`	I, static	96.9	4 847	independent probe, 133.0 m

Reporting rules. The map under test is `map/map_tiny.db`, created 8 September 2026 from `small_warehouse_static/small_warehouse_static_tinymap.world`. The reference occupancy was rasterised from the world file's collision geometry at 0.05 m per cell; the fitted placement is $(1.850, 8.600)$ m at $-89.90°$, measured once by `script/fit_map_pose.py`. Occupied-cell precision is the share of occupied map cells within 0.25 m of true structure; recall is the share of true *surface* cells within tolerance of an occupied cell, over the region the map observed and over the whole building; drivability stamps the 0.83×0.50 m footprint from `nav2_ackermann.yaml` at every ground-truth pose and counts collisions (false obstacles), unknown cells and clearance. All of it is produced by `script/validate_map.py` into `output/compare/rtab_map_vs_gt.png` (independent route) and `output/compare/rtab_map_vs_gt_selfcheck.png` (build run). The experimental unit is one database, $n = 1$.

Results. The tiny map is geometrically faithful: 96.6 % of the cells it marks occupied lie on real structure. Table 4.3 gives the three geometry scores and their populations; table 4.4 gives drivability against the lost independent route and against the build run; table 4.5 summarises the stored graph. Figure 4.1 is the evaluation overlay for the independent route and figure 4.2 the same database against its own build run.

Table 4.3: Occupied-cell geometry of `map/map_tiny.db` against `small_warehouse_static_tinymap.world` at 0.25 m tolerance.

Metric	Value	Population
Occupied-cell precision	96.6 %	11,634 occupied cells
Recall, region the map observed	89.5 %	observed surfaces
Recall, whole building	67.2 %	all surfaces in the world file

Table 4.4: Drivability of `map/map_tiny.db` under the true 0.83×0.50 m nav2 footprint, against two routes. "Independent" is `dataset_static_tiny_repeat`, which did not build the map and is lost. "Build run" is `dataset/dataset_map_tiny`, the run the map was built from. Geometry is identical in both cases because it does not depend on the route.

Metric	Independent	Build run	Meaning
Poses scored	2973	3032	
Clear and mapped	99.3 %	100.0 %	footprint free and observed
False obstacle	0.7 %	0.0 %	proven error
Unmapped under footprint	0.0 %	0.0 %	blind cells on the route
Clearance, median	1.07 m	1.01 m	to nearest obstacle
Clearance, 5th percentile	0.45 m	0.46 m	inflation radius is 0.80 m

Analysis. The gap between observed-surface recall and whole-building recall is route coverage, not mapping failure. The run never entered the side aisles, so about a third of the building was never seen by

Table 4.5: Stored contents of `map/map_tiny.db`, read in read-only mode. Link rows are stored per direction; the undirected count is the number of distinct node pairs.

Element	Rows	Undirected	Note
Nodes	30	–	ids 1–30, stamp span 59.3 s
Neighbour links	56	28	sequential odometry constraints
Global loop closures	5	3	pairs $(9,22)$, $(10,23)$, $(29,30)$
Gravity links	30	30	one per node
Local-space links	0	0	none recorded
Visual words	10,357	–	vocabulary size
Features	28,374	–	stored keypoints

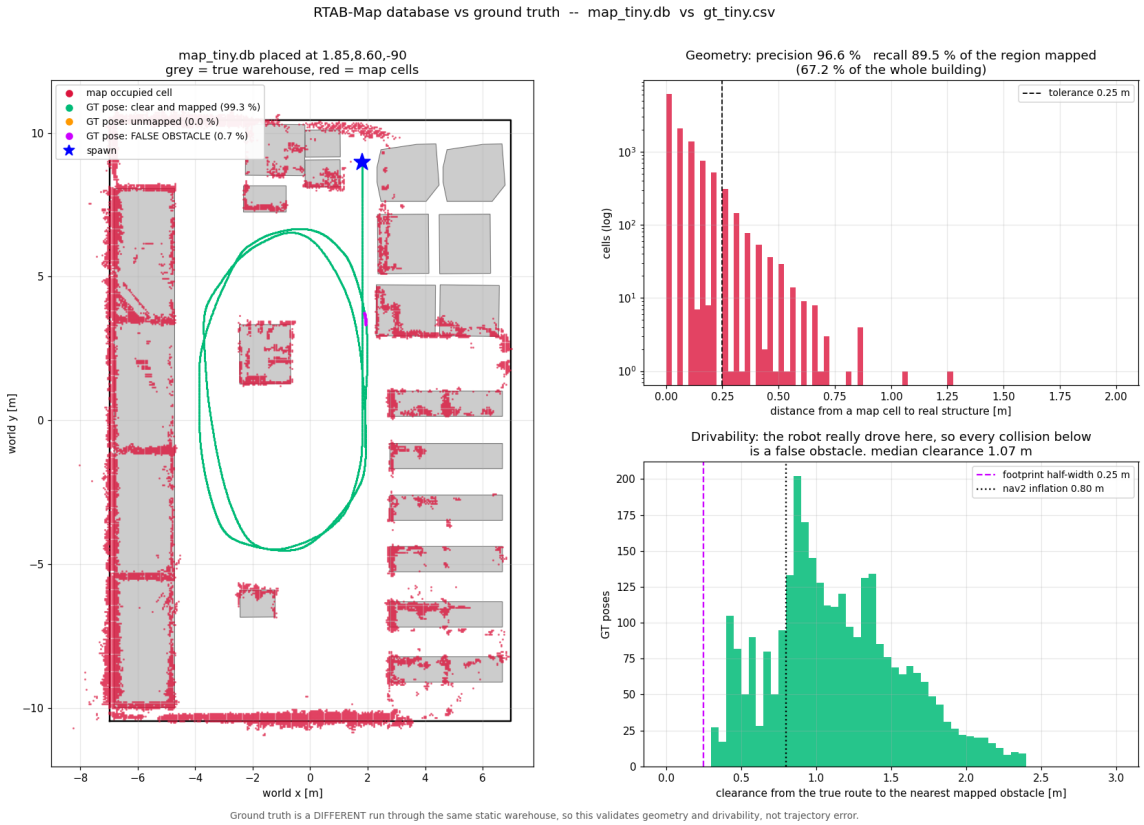


Figure 4.1: Dense-map evaluation of `map/map_tiny.db` against the independent route. Left: occupied cells (red) over the world file's collision geometry (grey), with the ground-truth route coloured by drivability and the spawn marked. Top right: distance from each occupied cell to real structure, with the 0.25 m tolerance marked. Bottom right: clearance from the route to the nearest mapped obstacle, against the footprint half-width and the nav2 inflation radius.

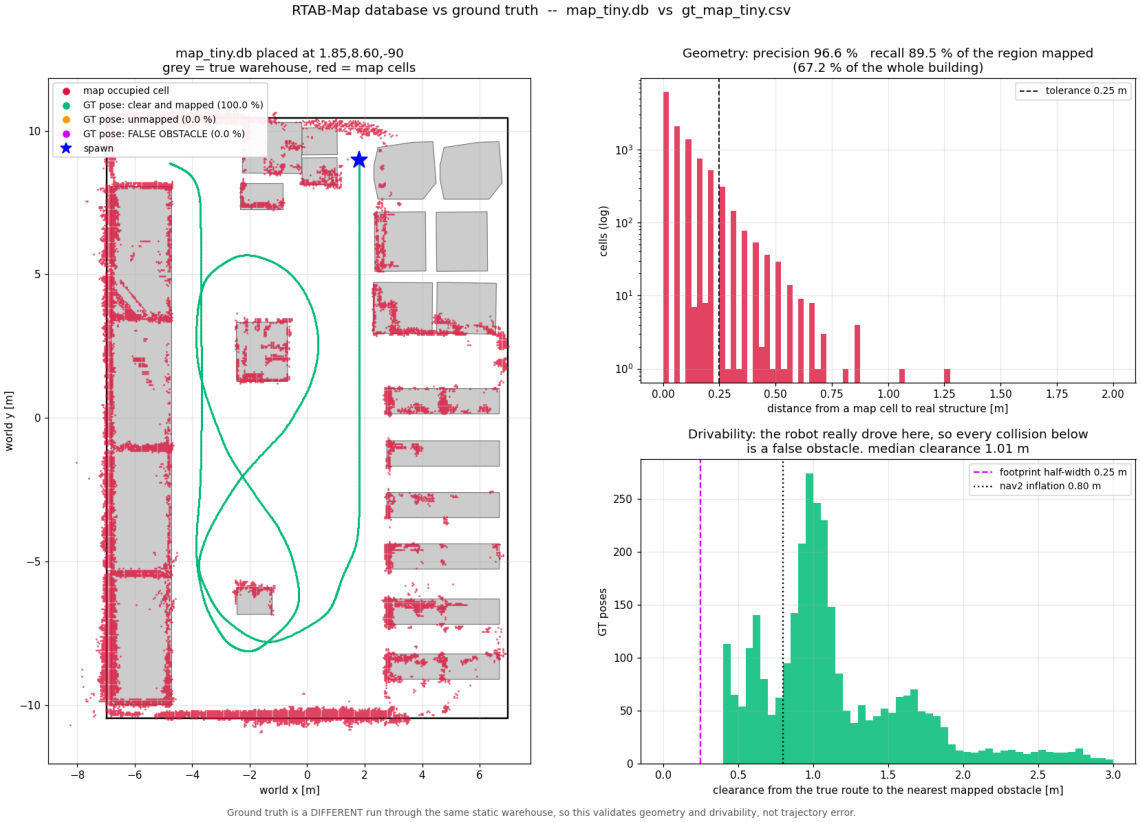


Figure 4.2: The same database scored against the run that built it, `dataset/dataset_map_tiny`. The route is a figure-eight rather than the oval of figure 4.1, and no pose on it collides with the map. The geometry panels are identical to figure 4.1. The footer claiming a different run is emitted unconditionally by the evaluator and is wrong for this figure.

any sensor; the first number describes mapping quality, the second where the robot went.

Placement is a measurement, not a tuned parameter. An independent prediction from the spawn pose of the IMU body frame gives $(1.800, 8.662)$ m, 0.080 m from the fitted value, and the fit beats the best rival orientation more than 5° away by a 60.9 % margin. Earlier assessments that judged this database unnavigable were made with the grid mis-seated, which is why placement is reported before any metric.

The two drivability columns measure the independence effect. Against the run that built it the map records no false obstacle at all, because it cannot contradict the trajectory it was constructed from; the 0.7 % that the independent route exposes is exactly what only an unseen route can reveal. The 0.0 % is a weaker question, not a better result, and is not a map-quality figure. Three measurements show the build run is the same run: its ground-truth poses at the stored node timestamps lie a median 0.340 m and at most 0.52 m from the nodes, the nodes lie a median 0.069 m from the route in space, and all thirty nodes fall inside the recording's sim-time interval.

Place recognition fired, but on a small graph. Two of the three closure pairs, $(9, 22)$ and $(10, 23)$, join nodes about half a lap apart and are genuine revisits; $(29, 30)$ joins consecutive nodes and is not evidence of place recognition.

Clearance couples to planner cost. The 5th-percentile clearance of 0.45 m sits below nav2's 0.80 m inflation radius on both routes, so nothing is blocked, but a planner would charge a high cost where the robot demonstrably drove.

Limitations, each stated once. *Lost reference:* the independent drivability route cannot be re-read, so that half of the result is a dated measurement. *Easy route:* one loop down the wide central aisle, so 0.7 % false obstacles is a floor, not a worst case. $n = 1$*:* one database on one world supports no uncertainty estimate and no claim about the large warehouse. *Residual distortion:* walls appear doubled and thickened and shelf rows bowed in the overlay; no rigid transform removes that, and it is the property that would limit navigation.

[Queried: reproducibility of the drivability half] The geometry result is reproducible: database and world file are both present. The drivability result is not: `dataset_static_tiny_repeat` is absent from `dataset/` and from the removable drive, the database stores an empty parameter set so its build configuration is unrecoverable, and the RTAB-Map engineering record's claim that the node stamps match no dataset is contradicted by the pose agreement above. The drivability figures stand as a dated measurement until an independent route is recorded and retained.

Large-warehouse attempt. The attempt **failed** before the back end was exercised. The progress update of 23 September 2026 records that the front end lost tracking in the repetitive bay structure of the large world, the estimated pose diverged from the ground-truth route within a few time steps of the loss, and the resulting map was not scorable; the cause was attributed to the front end under repeated structure, not to RTAB-Map. The mapping recording `dataset_map_01` survives in the registry. No run directory, placement fit, score or figure for this attempt is registered, so the result rests on the progress record alone and cannot be audited.

[Queried: unregistered databases] Four further databases exist beside the tiny map: `map/map_01.db` (0.77 MB), `map/map_02.db` (55.3 MB), `map/map_02_vins.db` (9.3 MB) and `map/map_03.db` (29.2 MB), all written on 21 September 2026. None is named in the progress record, the engineering record or any analysis output, so which recording, world and front end produced each is unknown. If any is the large warehouse, the attempt has retained an artefact after all and can be placed and scored with the unchanged evaluator before this slot is closed.

Realistic-warehouse maps. Two databases, one per layout, built on 26 October 2026 in the realistic worlds of Chapter 3 replace the unauditable large-warehouse attempt: each has a registered build recording, a fitted placement, a score and a figure. Both fail the quality criterion fixed before they were scored.

Reporting rules. The criterion, fixed in `script/eval_realistic_maps.py` before any map was placed, is occupied-cell precision $\geq 90\,\%$, observed-region recall $\geq 85\,\%$ and false obstacles $\leq 1\,\%$ of probe poses, each taken from the tiny map's result; whole-building recall is reported, not tested. Placement was measured by `script/fit_map_pose.py` with the realistic spawn $(5.250, 19.995, -90^\circ)$ and confirmed on the overlays before scoring. Scoring is the unchanged `script/validate_map.py` at 0.25 m tolerance with the 0.83×0.50 m footprint, each map probed by the static twin of its own layout. The databases store their full parameter set, unlike the tiny map: `Rtabmap/LoopThr` 1.0 and `RGBD/ProximityBySpace` false (`loop_closure:=false`, RTAB-Map record §7c). The grid was built to 8.0 m range (`Grid/RangeMax`; the record's measured default is 10.0) at 0.05 m cells and optimised with GTSAM (`Optimizer/Strategy` 2). Every number is under the experiment directory `exp4-realistic-maps` of `output/compare/simulation`: the scores in `realistic_map_metrics.csv`, one `validate_*.json` per map, and the graph inventory in `map_inventory.json`. The experimental unit is one database, $n = 1$ per layout. `map/map_realistic` is the build held until 26 October as `map_realistic_v2`; an earlier 114-node database of the same name was deleted that day, and its scores are retained under `superseded/` in the experiment directory.

Results. Table 4.6 gives geometry and table 4.7 drivability against the independent probes; table 4.8 summarises the stored graphs. Figure 4.3 is the placement overlay of the weaker fit and figures 4.4 and 4.5 the evaluation overlays.

Table 4.6: Occupied-cell geometry of the two realistic-warehouse databases against their world files at 0.25 m tolerance, beside the tiny map. Bold marks a value below the fixed criterion.

Database	Nodes	Occupied cells	Precision [%]	Recall observed [%]	Recall building [%]
`map_tiny`	30	11 634	96.6	89.5	67.2
`map_realistic`	165	26 728	**81.2**	**82.5**	55.7
`map_realistic_01`	185	25 440	**84.8**	**81.4**	54.5

Table 4.7: Drivability of the realistic-warehouse databases under the true 0.83×0.50 m nav2 footprint, against the independent probe of each layout; column heads drop the `map_` prefix. The tiny map's independent column is repeated for scale; its probe was one loop of the wide central aisle, whereas these probes thread the shelf aisles, so the two are not route-matched.

Metric	`tiny`	`realistic`	`realistic_01`
Probe poses	2 973	6 278	4 847
Clear and mapped [%]	99.3	96.2	94.9
False obstacle [%]	0.7	**3.8**	**4.4**
Unmapped under footprint [%]	0.0	2.4	0.7
Clearance, median [m]	1.07	1.96	2.16
Clearance, 5th percentile [m]	0.45	0.62	0.36

Analysis. Both maps fail, by similar margins, on every criterion: precision 81.2 and 84.8 against 90 %, observed recall 82.5 and 81.4 against 85 %, false obstacles 3.8 and 4.4 against 1 %. Agreement across two layouts and two independent builds makes this a property of the build rather than of one run.

The precision loss is mostly thickened structure, with a phantom share the tiny map did not have. Classifying the wrong cells (`script/exp4_insights.py`): 13.4 and 9.7 % of occupied cells are surfaces drawn too thick, within 0.50 m of true structure, and 5.2 and 5.3 % are phantoms further out, against 3.0 and 0.4 % on the tiny map. The thickening sits on the shelf rows the build route passed; the phantoms are the east and south walls, drawn about 0.5 m off their true line in both maps and far from any route. The false obstacles are not the thickened shelving: every blocked probe pose lies within 0.6 m of a

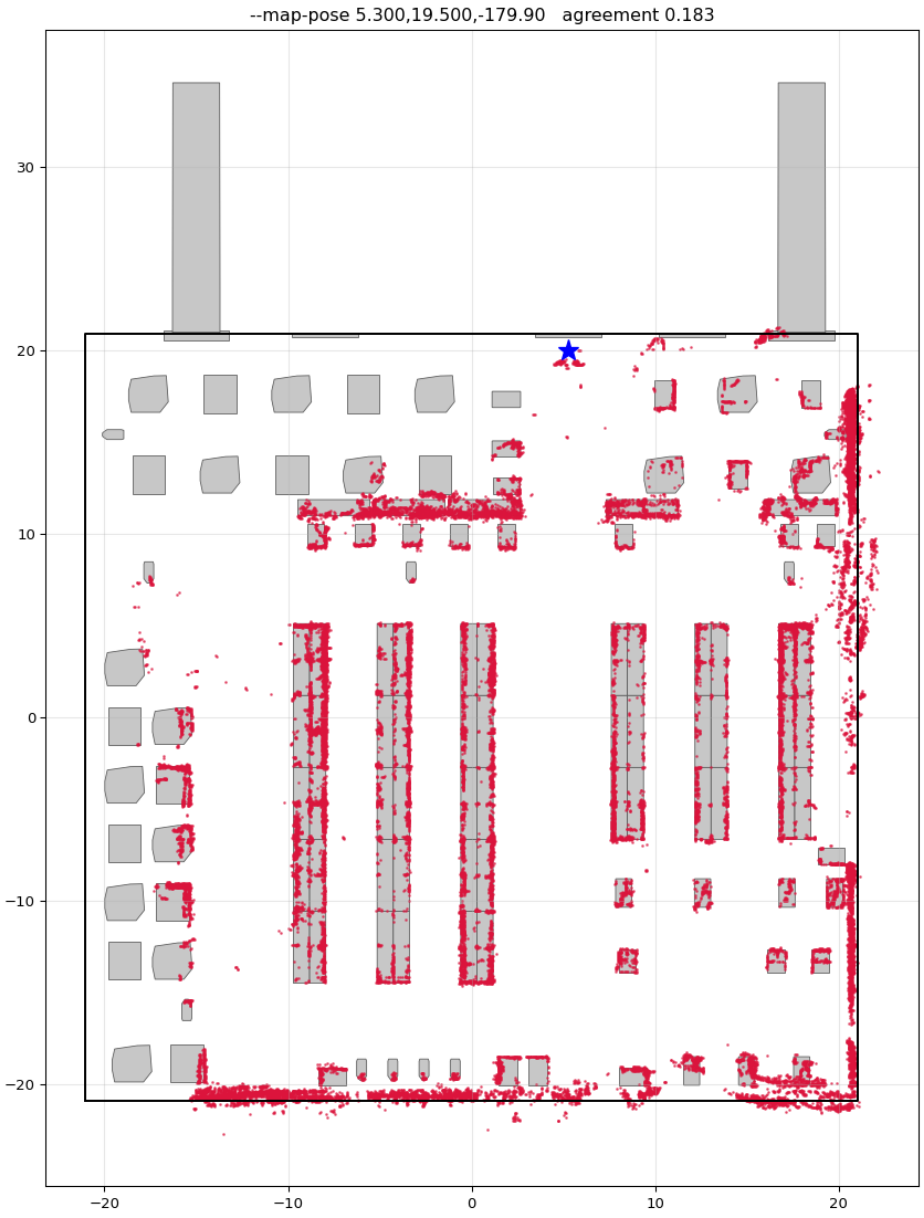


Figure 4.3: Placement overlay of `map/map_realistic`, the weaker of the two fits (agreement 0.183, 10.1 % margin over the rival orientation). Red occupied cells over the world file's obstacle footprints and wall. Shelf rows, the bay strip at $y \approx 11$ and all four walls coincide, so the fit is accepted on the overlay despite the thin margin; the `_01` fit agrees to within 0.7°.

Table 4.8: Stored contents of the realistic-warehouse databases, read in read-only mode. Neighbour links are stored per direction; closure and gravity rows once. “Spanning” closures join nodes more than three apart; none exists, by configuration.

Element	`map_realistic`	`map_realistic_01`
Nodes	165	185
Node stamp span [s]	285.7	314.9
Optimised path [m]	369.5	384.7
Neighbour links, rows / pairs	322 / 161	356 / 178
Global closures, rows	3	4
Closures spanning > 3 nodes	0	0
Gravity links	165	185
Visual words	55 802	63 276
Features	145 067	159 009
Database size [MB]	417	438
Fitted placement (x, y, θ)	$(5.30, 19.50, -179.9^\circ)$	$(5.15, 20.45, 179.4^\circ)$
Fit agreement, margin	0.183, 10.1 %	0.260, 66.1 %

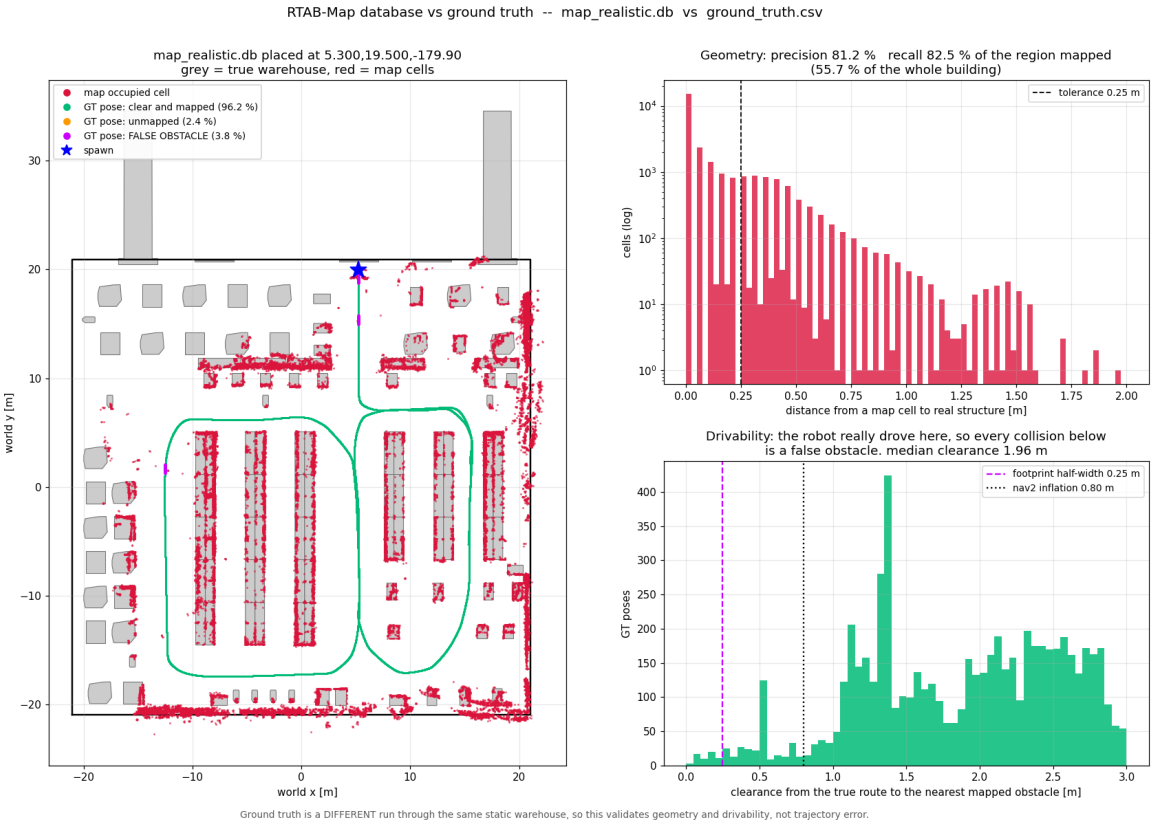


Figure 4.4: Dense-map evaluation of `map/map_realistic` against `dataset_allsensor_realistic_000`. Panels as in figure 4.1. The shelf rows are thickened toward the aisles; the false-obstacle poses sit on the first metres below the spawn, where the top wall is doubled, and where the probe hugs the left shelf column at $x \approx -12.5$.

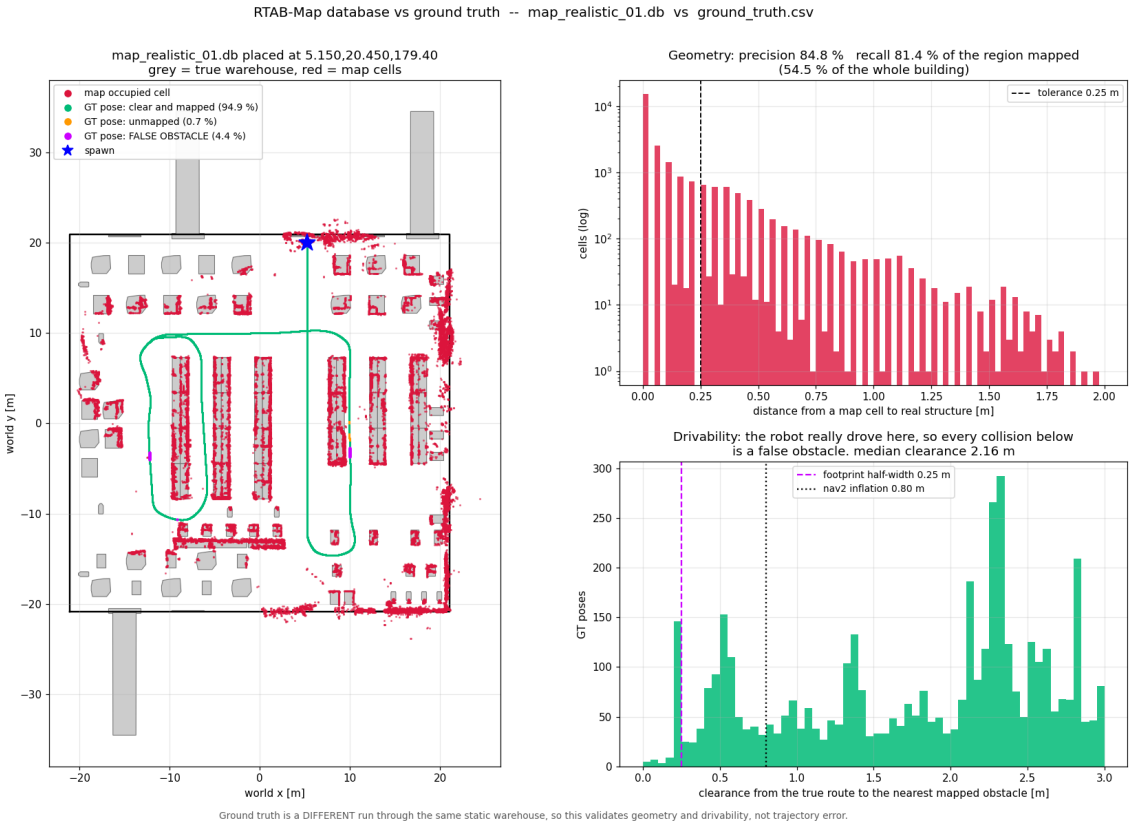


Figure 4.5: Dense-map evaluation of `map/map_realistic_01` against `dataset_allsensor_realistic_001`. The cleaner of the two in geometry. The clearance mode at exactly 0.25 m is the probe threading the one-aisle gap at $x \approx 10$ between two mapped shelf rows.

phantom cell, the largest blocked stretch of each map is the first metres from the spawn, and 96 and 63 % of the phantom cells under those stretches were first observed by one of the first five graph nodes, during the inertial initialisation transient.

These maps are an odometry picture by design, and the result should be read as one. Loop closure was disabled on both sides of the pipeline after the morning's A/B showed the front end's own closures corrupting an otherwise 0.1 m map (RTAB-Map record §7c), so no node was ever corrected against a revisited place; the stored graph confirms it, with no closure spanning more than three nodes. The thickening is therefore uncorrected drift along a 370–385-metre build route re-observing the same shelf from slightly different estimated places, which is exactly what this configuration was built to expose. It is not evidence about RTAB-Map's loop closing, which was not exercised.

Whole-building recall of 55.7 and 54.5 % is route coverage, as on the tiny map: the build routes loop the central block and the outer bays on the left and right were seen only in passing. Observed-region recall, the mapping figure, is 82.5 and 81.4 %.

Placement needed the overlay, and the spawn check did not apply. Both fits landed about 180° from the −90° spawn prediction, one with a thin margin, because the map frame was seated during the inertial initialisation transient rather than at the body's spawn pose: node 1 of each database carries a yaw of 57 and −89° and a height still settling over the first nodes, where the tiny map's node 1 is the identity. The overlays leave no doubt about the placement, and the two fits agree to 0.7°.

Limitations, each stated once. *Not route-matched:* the tiny probe was one loop of a wide aisle, these probes thread shelf aisles (5th-percentile clearance 0.36–0.62 against 0.45 m), so part of the false-obstacle gap is probe difficulty and the 1 % threshold was set against an easier route. *No attribution:* world, route, duration and path length vary together across the two, so no factor may be credited for a difference between them; the longer build, `_01`, is the cleaner. *n:* one database per layout supports no uncertainty estimate. *Front end unrecorded:* the database stores the back end's parameters but not which front end supplied the odometry chain.

[Queried: front end of the realistic builds] RTAB-Map took its pose from TF, so the odometry came from whichever front end was running; the database does not record which. The precision loss cannot be placed on the front end against the back end until the build logs name it.

Summary. Where tracking holds, the mapping pipeline is geometrically sound on the tiny map: occupied-cell precision 96.6 % and observed-surface recall 89.5 %, with whole-building recall of 67.2 % explained by route coverage; the drivability half is a dated measurement whose independent reference is lost. On the realistic warehouse two maps built without loop closure are auditable end to end and fail the same criterion by similar margins: precision 81.2 and 84.8 and false obstacles 3.8 and 4.4 % against independent probes. Their loss is thickened shelving, displaced walls and spawn-area phantoms from uncorrected drift over 370–385-metre routes, which the closure-free build was configured to expose. The earlier large-warehouse attempt that lost tracking remains unauditable.

Future plan. The two realistic maps are the odometry baseline that a corrected map is to be compared against with the unchanged evaluator: re-enable closure with a translation bound on accepted corrections, or merge the closure-free sub-maps through Atlas, and rescore. Name the front end in every build manifest. Record the tiny-warehouse independent route again and retain it. Identify and score the 21 September databases, or let the realistic maps stand as the large-warehouse result.

4.4.2 Real robot

Status: planned. No RTAB-Map database has been built from real imagery, and no reference geometry of the site exists to score one against.

Reason. The geometry metric scores occupied cells against known collision geometry, and the placement step needs that geometry; the mock warehouse has no such model. The drivability metric needs a tracked route from a run other than the mapping run. Neither input exists at the cutoff.

Next action. Survey the mock warehouse (walls, stacks, walkway) into a reference occupancy grid at 0.05 m, build a database from a real mapping run, and score it with the unchanged evaluator, using the optical tracker route from a second run for drivability.